EL NIÑO ACTION BRIEFING · 20 OCTOBER 2026

Sants-Badal, Barcelona

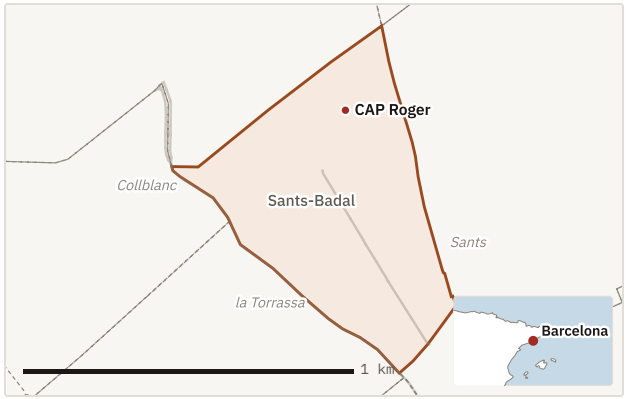
The winter threat here is flu and cold homes, not El Niño heat. The first job is to get every eligible household on the bono social, on the regulated tariff, before 31 December.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Cataluña (Climate Impact Lab region)	94 in Cataluña; about 0.3 in the barri (our scaling)	Sep 2026 – Feb 2027 (6-month); 62 of the 94 fell in September	Flu and cold homes; flash floods second

18/18 recommendations checked (7 actions, 7 warnings, and 4 funding placements). 10 corrected, 1 unverified, 2 contraindicated, 5 confirmed as written. The two contraindicated proposals, pre-buying food stock and a top-up to the city's energy advice points, were dropped or moved to Do not do this. Exposure figure: Climate Impact Lab / EPIC, Sep–Feb heat-death map (21 September 2026), pre-peer-review, heat only. The barri figure is our population scaling, not an Impact Lab finding. **The funding target fell by half in verification**, from €125,000 to €62,500 (US$70,156): the city's energy advice points cannot take outside money, and heating cash duplicated municipal aid.

Read this first: why this is not an "El Niño winter" briefing. A strong El Niño is under way (NOAA, 8 October 2026: Niño-3.4 at +2.1 °C, still strengthening), but its measured effect on Barcelona's winter is weak. The published signal in eastern Spain (Rodó, Baert and Comín, 1997) shows up mainly in the following spring, and its strength and sign change from decade to decade. Catalan downpours follow the Western Mediterranean Oscillation instead. Most of the projected heat deaths for Cataluña fell in September, which has

EXECUTIVE SUMMARY — AT A GLANCE



41.37°N 2.13°E, Spain · © OpenStreetMap contributors (ODbL); Natural Earth

- **The risk.** Flu and cold homes kill here every winter; this year a court ruling ended Catalonia's block on power and gas cut-offs.
- **First action.** The city's energy advice points and volunteers enrol eligible households on the bono social, on the regulated tariff, by 31 December.
- **The money.** €17,500 floor, €62,500 target, €80,000 ceiling (US$19,644 / 70,156 / 89,800); €10,000 for flu-vaccination outreach buys the most.
- **Do not.** Do not give "stay warm" advice without carbon monoxide and fire safety: braziers, charcoal, and candles kill in cut-off homes.*

What to do	**7** actions	p. 2
Who does what	RACI chart	p. 8
Where money does the most	funding	p. 8
Do not do this	**7** items	p. 10
What has happened here	the record	p. 12
How this was produced	methodology	p. 14
References	sources	p. 16

* Suggested formulation: check with a health worker before sharing.

passed. So this briefing covers what kills people here every winter, whatever El Niño does: flu and cold homes, made worse this year by a court ruling that removed Catalonia's main block on electricity and gas cut-offs.

Scope. The request named a single street, which is finer than any data, so this briefing covers the barri that contains it: Sants-Badal, in the Sants-Montjuïc district (about 24,200 residents, January 2022). The street runs about 250 metres east of Carrer de la Riera Blanca, the municipal line with L'Hospitalet de Llobregat. Neighbours across that line in Collblanc and La Torrassa are served by a different city council, so several actions name both.

WHAT THIS PLACE SHOULD DO — RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action. An action several lenses reached independently is the stronger bet, but only where its evidence came from somewhere like here.

1. Enrol every eligible household on the bono social, on the regulated tariff, before 31 December

PARTIALLY CONFIRMED · ECONOMIST · FRONTLINE · SKEPTIC · PUBLIC HEALTH · PHILANTHROPIST · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Ajuntament de Barcelona, energy advice points (PAE), with neighbourhood volunteers; Creu Roja on the L'Hospitalet side	31 December 2026	Within the PAE contract; €7,500 (US$8,419) for the L'Hospitalet side	Power and gas cut-offs this winter; loss of the 2027 heating grant

On 24 June 2026 the Constitutional Court's ruling STC 42/2026 struck down articles 6.4 and 9.4 of Catalan Law 24/2015 for electricity and gas. Those articles made suppliers ask social services before cutting off a household. For power and gas, the protection left is state-level: households on the bono social have a supply guarantee to 31 December 2026 (RDL 7/2026), and those at risk of exclusion cannot be cut off at all. The bono also carries a 42.5% or 57.5% discount, and holding it on 31 December is what makes a household eligible for

* Suggested formulation: check with a health worker before sharing.

the next Bono Social Térmico heating payment (€90.86–400.69 in Catalonia's 2025 round). Take-up is low: 24.5% of eligible households in Spain in 2022, and 9.2% of pensioners on the minimum pension (EsadeEcPol). Enrolment must include switching to the regulated PVPC tariff with a reference supplier, which a form alone does not do.

Transferability: no trial has measured deaths averted by bono enrolment. The best evidence that help with the paperwork beats information alone is a US trial on food-aid uptake among older people (Finkelstein and Notowidigdo, 2019: about 6 points from information, about 12 with application help). The PAEs stopped 44,449 cut-offs in 2023. Water cut-offs are outside the ruling: the social-services check still applies to water.

2. Vaccinate housebound people aged 75 and over at home against flu and COVID, and escort the rest, by 18 December

PARTIALLY CONFIRMED | PUBLIC HEALTH · PHILANTHROPIST · EMERGENCY MANAGER | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Primary care team of ABS 3I Sants-Badal (CAP Roger), Catalan Health Institute; Radars and Amics de la Gent Gran volunteers	**18 December 2026**	**Vaccine free; outreach €10,000 (US$11,225)**	**Flu deaths at the late-December peak**

The 2026–27 campaign opened on 5 October for people aged 80 and over and care-home residents, and opens on 26 October for other people over 60; the push runs to 18 December. In a peer-reviewed test-negative study of people over 65 in Ontario, vaccination cut deaths within 30 days of confirmed flu by 20% (95% CI 7–30%; Chung et al., Clinical Infectious Diseases 2021). In Catalonia's 2024–25 season, 75% of flu patients in intensive care were unvaccinated. Volunteers phone, book, and walk people to CAP Roger; the CAP team vaccinates those who cannot leave home.

* Suggested formulation: check with a health worker before sharing.

Transferability: the effect size comes from Canada, a colder country, though the population is the same age and the virus the same. Coverage in this ABS is not published; Catalan coverage in 2025–26 was 39.6% across target groups. Adult RSV vaccine is not offered to people living at home in Catalonia, so do not promise it.

3. Ask the Spanish Government to extend the bono social discounts and supply guarantee past 31 December

CONFIRMED · ECONOMIST · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Ajuntament de Barcelona and Generalitat, to the Spanish Government	Before 31 December 2026	No budget needed to ask	Discounts dropping and the cut-off guarantee lapsing in the coldest weeks

RDL 7/2026 sets the higher discounts and the supply guarantee only to 31 December 2026. On 1 January 2027 the discounts fall back to 35% and 50%. The decree of 29 September (RDL 25/2026) extended other crisis measures but not these, and no extension has been announced. The policy cliff falls in the same weeks as the winter flu peak.

Transferability: this is a national decision, and every Spanish city faces the same date. It is listed here because the city and the Generalitat are the bodies that can ask, and because action 1 is worth less if the guarantee lapses.

4. Visit people over 75 who live alone: check heating, cut-off notices, and heaters, and leave a battery lamp

PARTIALLY CONFIRMED · EMERGENCY MANAGER · FRONTLINE · SKEPTIC · HISTORIAN · PUBLIC HEALTH · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Serveis Socials Sants-Montjuïc with the Radars network; L'Hospitalet social services for Collblanc–La Torrassa	1 December 2026	Staff and volunteer time; packs €45,000 (US$50,513) if a distributor is confirmed	Deaths behind a closed door: cold, fire, and carbon monoxide

* Suggested formulation: check with a health worker before sharing.

In November 2016 in Reus, Rosa, aged 81, died in a fire in her flat after two months without electricity; the supplier had cut her off without telling social services. Carbon monoxide from braziers, stoves, and heaters kills an estimated 50–125 people a year in Spain (SEPAR and health-ministry estimates). The visit checks for a cut-off notice, a working heater, and an unsafe one, gives a battery lamp instead of candles, and refers the household to action 1. Ask for the last electricity bill rather than "are you cold?": people show a bill more readily than admit to the cold. Use the existing Radars shop network, with a one-page sheet in Catalan, Spanish, Italian, Urdu, and Chinese (the barri's largest foreign groups in 2022 were Italian, Pakistani, and Chinese).*

Transferability: there is no direct evidence that carbon monoxide alarms reduce deaths, so an alarm is a backstop to the advice, not the action. Radars has no published mortality outcome. Its 2025–26 activity in this district was not confirmed; check with the district office. In 2022, about 2,756 residents were 75 or over and 31.7% of them lived alone (a figure from the 2023 municipal profile that could not be re-opened).

5. Ask the Generalitat to approve the decree for the basic-supplies solidarity fund

CONFIRMED | ECONOMIST · HISTORIAN | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Generalitat de Catalunya (social rights)	Before the winter peak	None to ask	Arrears since 1 January 2026 turning into cut-offs

The Endesa agreement that split vulnerable households' debts 50/50 between the company and the public authorities expired on 31 December 2025 and was not renewed; Naturgy and Iberdrola never signed. Arrears since January 2026 have no forgiveness mechanism. A draft decree for a Fons d'Atenció Solidària de Subministraments Bàsics went to public consultation but had not been published as of 20 October.

* Suggested formulation: check with a health worker before sharing.

Transferability: a Catalonia-wide fix. The barri cannot do it, but its social services carry the cost of not having it.

6. One flood list for both sides of the municipal line: ground-floor homes, door-knock teams, and messages in five languages

PARTIALLY CONFIRMED | EMERGENCY MANAGER · HISTORIAN · SKEPTIC | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Barcelona and L'Hospitalet civil protection; Barcelona Cicle de l'Aigua (BCASA) for the drains	15 November 2026 for the list; drains before the next red warning	Not costed; staff time and existing maintenance budgets	Deaths in flooded ground floors and basements

On 3–4 October 2026 Meteocat recorded the wettest Saturday in at least 20 years, with 100–200 litres per square metre across the Barcelonès. Two people died in Catalonia and one was missing. In L'Hospitalet most calls were about flooded ground floors. Barcelona and L'Hospitalet should keep one shared list of ground-floor and basement dwellings near the Riera Blanca line, name the teams who knock on those doors on a red warning, and clean the gullies along Riera Blanca, Collblanc, and Badal before the next heavy rain. Only the Generalitat can send an ES-Alert, and it goes out in Catalan, Spanish, or English, so Urdu, Chinese, and Tagalog versions must travel by door and by WhatsApp.

* Suggested formulation: check with a health worker before sharing.

Transferability: no source names Sants-Badal, Collblanc, or La Torrassa among the streets hit on 4 October, and no street-level incident log is public. The Riera Blanca is a buried stream on the municipal line; Collblanc flooded in August 1981, when people were rescued from cars. Cleaning drains helps in ordinary storms; on a 100–200 mm night the sewers overflow anyway.

7. Top up the Targeta Barcelona Solidària food card for December to February

PARTIALLY CONFIRMED | FOOD SUPPLY CHAIN | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Institut Municipal de Serveis Socials (IMSS), Ajuntament de Barcelona	30 November 2026	About €300,000 (US$336,750) for 1,000 Sants-Montjuïc households at €100 a month (our estimate); unfunded	Skipped meals as oil and cereal prices pass through

The El Niño food signal here is weak and late. The pressure now is fuel and freight: the FAO Food Price Index stood at 136.0 in September 2026, with vegetable oils up 18.3% and cereals up 17.2% on a year earlier, and Spain's inflation was 4.9%. El Niño's effect on palm oil and rice mostly arrives in 2027. Neighbourhood shops will stay stocked; household budgets will not. The city already runs a rechargeable food card, so cash through it beats food parcels. People renting a room without a kitchen cannot use a card, and need meals; the only public canteen confirmed in the district is on Avinguda del Paral·lel, about 2 km away, and its spare capacity is not published.

Transferability: the card is Barcelona's own channel. The current number of cards in use was not found, and the €300,000 is our arithmetic, not a budget line. Watch the FAO vegetable oil index (monthly; next releases 6 November and 4 December) and INE's food price index (monthly, mid-month).

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Ajuntament de Barcelona	Ajuntament de L'Hospitalet	CAP Roger (ICS)	Generalitat	Spanish Government	Community groups
1. Bono social enrolment	**A/R**	**R**	**I**	**I**	**I**	**R**
2. Flu and COVID vaccination	**C**	**I**	**A/R**	**C**	**–**	**R**
3. Extend the bono past 31 Dec	**R**	**C**	**–**	**R**	**A**	**C**
4. Home visits to over-75s alone	**A/R**	**R**	**C**	**–**	**–**	**R**
5. Supplies solidarity fund decree	**C**	**C**	**–**	**A/R**	**I**	**C**
6. Cross-border flood list	**A/R**	**R**	**–**	**C**	**–**	**I**
7. Food card top-up	**A/R**	**–**	**–**	**–**	**–**	**C**

What the chart shows: Barcelona city council is accountable for four of the seven rows. Rows 3 and 5 need a decision rather than money, and they are the ones the barri cannot make itself. L'Hospitalet is responsible on three rows because the street sits one block from its boundary.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per euro. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR

€17,500 (US$19,644)

TARGET

€62,500 (US$70,156)

CEILING

€80,000 (US$89,800)

LARGEST UNCERTAINTY

Whether anyone can distribute the safety packs; and flu coverage in this ABS, which is unpublished

Floor: €10,000 for outreach (250 contacts at €40) plus €7,500 for L'Hospitalet enrolment (150 households at €50) = €17,500. Target: the floor plus €45,000 for 300 safety packs at €150 = €62,500, released only once a distributor shows it can deliver. Ceiling: outreach at €20,000 (500 contacts, the most a city-wide volunteer group supporting about 450 people can plausibly add) plus enrolment at €15,000 (300 households, near Creu Roja's 2025 L'Hospitalet volume of 263) plus the packs = €80,000. The unit costs of €40, €50, and €150 are working assumptions, not published figures. **The target fell by half in verification**, from €125,000: the PAE placement was re-routed, because a city-contracted service cannot take outside money, and €45,000 of heating cash was dropped because municipal urgent aid already covers it. Even the target is

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unlikely to avert more than about one death in this barri. It is worth funding because it buys timing: people vaccinated by 18 December and enrolled by 31 December.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Fundació Amics de la Gent Gran	Phone, book, and escort over-75s to vaccination (action 2)	€10,000 (US$11,225)	Direct grant	250 contacts at €40 (assumption). Effect: 20% fewer deaths after confirmed flu (Chung 2021, Ontario). The vaccine itself is publicly funded.	Partially confirmed
Creu Roja, L'Hospitalet de Llobregat	Bono social enrolment for Collblanc–La Torrassa households (action 1)	€7,500 (US$8,419)	Direct grant	150 households at €50 (assumption). Precedent: Creu Roja advised 263 people in L'Hospitalet in 2025, and 117 of 500 advised obtained the bono.	Partially confirmed
Distributor to be confirmed (Ecoserveis named, not shown to distribute kits)	Safety packs: battery lamp, thermal bedding, carbon monoxide alarm (action 4)	€45,000 (US$50,513)	Pre-positioned stock	300 packs at €150 (assumption). No mortality evidence for the contents; the alarm is a backstop.	Unverified

The funder's one thing: the cheapest protection here is paperwork, not cash, and its deadline is 31 December. Across the street, Collblanc–La Torrassa is denser and poorer than Sants-Badal, and Barcelona bodies cannot spend there, so the L'Hospitalet placement probably buys more per euro than its rank suggests. Dropped in verification: the PAE top-up (it cannot take outside money; enrolment there is already free) and heating cash (municipal urgent social aid already pays it).

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives, or waste the window, here.

Do not give "stay warm" advice on its own *

CONTRAINDICATED

In homes whose power has been cut, people warm themselves with braziers, charcoal, butane heaters, and candles. That is how the Reus death happened in 2016, and carbon monoxide kills an estimated 50–125 people a year in Spain. Generators killed at least three people during the April 2025 blackout.

Instead: pair any warmth advice with the safety message (no braziers or charcoal indoors, ventilate gas heaters, no candles), a battery lamp, and a referral to stop the cut-off (actions 1 and 4).

Do not pre-buy food stock in November

CONTRAINDICATED

Prices are already high (FAO cereals +17.2% and vegetable oils +18.3% year on year) and goods are not short. El Niño's effect on rice and palm oil mostly lands in 2027, so buying now locks in high prices and carries the stock through the peak. The Gran Recapte food drive already brings in donated stock in late November.

* Suggested formulation: check with a health worker before sharing.

Instead: put the money on the existing food card (action 7), and fund meals for people who cannot cook.

Do not call this an "El Niño winter" or quote an El Niño death toll for the barri

CONTRAINDICATED

The El Niño link to Barcelona's winter is weak and unstable, and the only quantified projection covers heat, mostly in September. Scaled to the barri it is about 0.3 deaths. Selling cold and flu work as El Niño work invites the reply "it was a mild winter after all" and loses the budget for next year.

Instead: present it as winter and flood readiness that pays off whatever El Niño does.

Do not answer the cold with "go to a climate shelter"

CONTRAINDICATED

Barcelona's climate-shelter network is large (about 540 sites in summer 2026), but no independent evaluation of who uses it exists, and the most-cited usage figures are opposition claims from 2023. No 2025–26 winter list was found, and no shelter was confirmed in Sants-Badal. Shelters do not reach the frail person who cannot get downstairs, and do not fix a cold home overnight. A Barcelona shelter may also turn away a resident from across the street in L'Hospitalet.

Instead: check a named site's winter hours and step-free access before referring anyone, and send someone to the person who cannot go.

Do not run a leaflet campaign for the bono social

CONTRAINDICATED

Three in four eligible households, and nine in ten minimum-pension pensioners, already go without it, and the information exists. Application also needs a switch to the regulated tariff, which a leaflet cannot do.

* Suggested formulation: check with a health worker before sharing.

Instead: fill in the application and the tariff switch with the person, at the door or at the energy advice point (action 1).

Do not rely on the ES-Alert alone for flash floods

CONTRAINDICATED

In the Valencia DANA of 29 October 2024, the first ES-Alert went out at 20:11, about three hours after the option was first raised, and the investigating judge called it late and wrong. In Catalonia too the alert depends on a Generalitat decision, and it does not come in Urdu, Chinese, or Tagalog.

Instead: a named door-knock list for ground floors and multilingual messages that go out on Meteocat's warning, not on the alert (action 6).

Do not sell insulation or façade work as lifesaving this winter

CONTRAINDICATED

In Barcelona's own study of façade retrofits in public housing (Peralta et al., 2017), the link between cold and deaths did not fall in men; in one time window it rose (interaction RR 2.23, 95% CI 1.14–4.36). Retrofits cannot be finished before the peak in any case.

Instead: spend this winter's money on enrolment, vaccination, and visits; plan retrofits on their own merits.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1962	Vallès flood, 25 September, north of Barcelona	617 dead by the official count; estimates of 800–1,000	Remembered as Catalonia's worst flood; not specific to Sants
1981	Downpour on L'Hospitalet, 21 August; people rescued from cars at Collblanc	Not recorded	The sewer network was judged too small. Barcelona later built storm tanks; the nearest, Joan Miró (55,000 m³), opened in 2003

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
2008	Drought; Barcelona shipped water in by tanker	None recorded	A desalination plant opened in 2009
2010	Wet snow, 8 March; 33 high-voltage towers fell	About 200,000 homes and businesses lost power	Not researched for this place
2016	Rosa, 81, died in a fire in Reus after two months without electricity	1 death	Enforcement of Law 24/2015; the supplier was fined. In 2026 the Constitutional Court struck the key articles down for power and gas
2022	Summer heat in Catalonia	2,705 excess deaths from all causes, June–August; 582–1,772 attributed to heat	The climate-shelter network grew to about 540 sites by 2026
2024	Drought emergency, declared 1 February during the 2023–24 El Niño; reservoirs at 14%	None recorded	Per-person water caps, lifted for most of Catalonia in May 2024
2026	Storm of 3–4 October: 100–200 l/m² across the Barcelonès; Sants station partly flooded	2 dead and 1 missing in Catalonia, none in Barcelona or L'Hospitalet	Too recent to say

The recurrence nobody has noticed: Carrer de la Riera Blanca is both a buried stream and a municipal boundary, and the record keeps showing this corner falling between two councils. Shacks grew there because nobody planned it, Barcelona's storm tanks stop at the line, and social services, refuges, and energy advice points serve one side or the other. The risk here is less El Niño than jurisdiction.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Catalan Law 24/2015 still blocks power cut-offs** versus **the Constitutional Court struck it down**. The Economist, Historian, and Public Health lenses relied on the law; the Frontline lens said it had gone. Verification settled it from the BOE: STC 42/2026 annulled articles 6.4

* Suggested formulation: check with a health worker before sharing.

and 9.4 for electricity and gas, while water stays protected. Only the original wording was ruled on, and suppliers are treating the amended text as gone too; a court challenge to the amended text would settle it.

- **Flooding is the first priority** (Emergency Manager) versus **no flood record exists for these streets** (Frontline). The October storm was real, but no source places damage in Sants-Badal or Collblanc. A street-level incident log from the two councils would settle it; it is not public.
- **Heating cash for older people** (Philanthropist) versus **efficiency and enrolment beat subsidies** (the Philanthropist's own NICE citation, which could not be opened). Verification dropped the cash because municipal urgent aid already pays it.

WHAT NOBODY COVERED — BLIND SPOT

Housing. No lens measured how many people in the barri rent a room in an overcrowded flat, live without a contract, or are undocumented. They are the households least likely to qualify for the bono social, to open the door to officials, or to have a kitchen. The eviction moratorium ended in March 2026. If they are numerous, actions 1 and 4 reach the wrong half of the street, and the tenant and housing groups (the Grup d'Habitatge de Sants, the Sindicat de Llogateres, the PAH) become the delivery channel rather than social services. Nor did any lens cover care homes, where flu outbreaks kill most.

HOW TO READ THIS — METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab region scale, here all of Cataluña (about 8 million people). They are not neighbourhood figures. The El Niño projection covers *heat only*; cold, flu, and flood are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED — METHODOLOGY

- **Scope.** The requester typed a single street. A street is finer than any data, so the location was widened to the barri that contains it, Sants-Badal (Sants-Montjuïc district), with the L'Hospitalet side of the municipal line named where it changes the action. Climate Impact Lab

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region: Cataluña, as its Sep–Feb heat-death map gives it. Mortality figures exist only at that scale; sub-city detail is used for hazard exposure and action design, not for death counts.

- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline. The regional figures come from the Sep–Feb heat-death map credited to EPIC (21 September 2026): Cataluña 94 (September 62, October 7, November 9, December 8, January 3, February 5) and Spain 480. Scaled by population, 94 × 24,200 / about 8,000,000 ≈ 0.3 for the barri; that is our arithmetic, not an Impact Lab finding. It covers heat only and is never added to the Impact Map's long-run projections. An earlier briefing for Alcoy (10 October) said no Spain figure existed; that was wrong.
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Six verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed. The tally counts the 7 actions, the 7 "Do not" items, and the 4 proposed placements.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted by reading the data embedded in the inventory page on 20 October 2026. It has no entries for Spain. Its nearest Mediterranean precedents are Italian heat programmes, which do not cover cold, flu, energy cut-offs, or flood warning.
- **Money.** Costs are shown in euros with US dollars in parentheses, at €1 = US$1.1225 (European Central Bank euro reference rate, 19 October 2026).

* Suggested formulation: check with a health worker before sharing.

- **Health wording.** Passages where a wording error could hurt someone (carbon monoxide, braziers, candles, and fire safety in homes without power) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Unit costs and the food-card figure are labelled as our estimates. Every source is listed below.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in Sants-Badal or L'Hospitalet was interviewed. Several Generalitat and Canal Salut pages returned 403 errors, so some campaign and coverage figures come from press summaries. The barri's 2023 statistical profile could not be re-opened, so the age and living-alone figures are from 2019–2022. No flu-vaccination coverage, heating prevalence, or winter excess mortality is published for this barri. Street-level flood incidents for 3–4 October are not public. The court order in the Valencia DANA case was read only through press reports.

References · status after verification in brackets

Premise and exposure · NOAA CPC ENSO discussion, 8 Oct 2026 [confirmed] · Shaman and Tziperman 2011, citing Rodó et al. 1997 [confirmed] · López-Bustins, WeMO and Catalan rainfall [confirmed] · Sep–Feb heat-death map (EPIC) [read directly] · Healy 2003, excess winter mortality in Europe [confirmed]

Energy and money · STC 42/2026, BOE 24 Jun 2026 [confirmed] · RDL 7/2026 [confirmed] · RDL 25/2026 [confirmed, no bono extension] · Bo social tèrmic FAQ [confirmed] · EsadeEcPol on bono take-up [confirmed] · Finkelstein and Notowidigdo 2019 [confirmed] · Barcelona PAE results [confirmed] · End of the Endesa agreement [confirmed] · Creu Roja Cita Bono Social, 2025 [confirmed]

Health · Chung et al. 2021, Clin Infect Dis [confirmed, CI corrected] · Canal Salut, autumn vaccination [confirmed via summary] · ASPB, ABS 3I Sants-Badal at CAP Roger [confirmed] · Carbon monoxide deaths, health-ministry estimate [estimate] · Reus, November 2016 [confirmed] · Peralta et al. 2017, façade retrofits and cold mortality [confirmed] · Sants-Montjuïc community resources for older people (Radars) [confirmed, 2025–26 activity not] · Sants-Badal barri profile [2019 data] · Catalonia summer 2022 excess deaths [confirmed]

Flood · RTVC/EFE, storm of 3–4 Oct 2026 [confirmed, figures corrected] · L'Hospitalet after the storm [confirmed] · How Catalonia's ES-Alert works [confirmed] · Valencia DANA alert timing [press report of the court order] · Joan Miró storm tank [confirmed] · 1962 Vallès floods [secondary]

Food · FAO Food Price Index [confirmed] · INE CPI, September 2026 [confirmed] · Targeta Barcelona Solidària [confirmed, current count not found] · Mercabarna flood protocol [confirmed]

Shelters · Barcelona passes 500 climate shelters [confirmed] · Opposition criticism of shelter use [opposition claims, 2023]

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory

Locator: © OpenStreetMap contributors (ODbL); Natural Earth.

* Suggested formulation: check with a health worker before sharing.

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